

Digital footprints: Incorporating crowdsourced geographic information for protected area management

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A B S T R A C T

Biodiversity loss driven by anthropogenic pressures highlights the importance of conservation efforts in protected areas globally. Protected areas are also locations providing myriad ecosystem services, including recreation and tourism. Advancements in mobile and web technologies have expanded the capabilities and accessibility of crowdsourced spatial content increasingly leveraged for research. This study explores the use of crowdsourced geographic information to model, at varying temporal scales, spatial patterns of visitor use and identify factors contributing to distribution patterns in a dynamic landscape, Hawaii Volcanoes National Park (Hawaii, USA). Specifically, this study integrated geotagged photo metadata publicly shared on Flickr with raster data about infrastructure and natural environment using MaxEnt modelling. Infrastructure designated for visitor use (i.e., roads, trails) contributed most to models of visitor distribution for all years and seasons. During the spring months, elevation was also a top contributing variable to the model. Crowdsourced data provided empirical assessments of covariates associated with visitor distributions, highlighting how changes in infrastructure and environmental factors may influence visitor use, and therefore resource pressures, to help researchers, managers, and planners with efforts to mitigate negative impacts.

1. Introduction

Environmental changes and biodiversity loss attributed to human activities increased more rapidly in the last century than during any other time in recorded history (Millennium Ecosystem Assessment, 2005; Pimm et al., 2014). This unprecedented rate and scale of human impact and ecological footprint ushered proposals to name a new geological epoch, the Anthropocene (Malhi, 2017; Steffen et al., 2011). The extent and complexity of human-environment interactions requires innovative approaches to address multidimensional human-environment interactions, fostering a paradigm shift in how such interactions are measured, monitored, and addressed (Harden et al., 2014). Dynamic feedback and interacting effects leading to increased complexity have resulted in studies analyzing such interactions as coupled systems (An & López-Carr, 2012).

Geography and Geographic Information Science (GIScience) are central to exploring, understanding, and modelling the human footprint (Sanderson et al., 2002) and human-environment interactions (Zvoleff & An, 2014), coupling social and ecological datasets through geographic location for integrated analysis. Increasing amounts, access, and accuracy of spatial data available for integrated analysis has fostered a paradigm shift in GIScience (Elwood, Goodchild, & Sui, 2012).

Specifically, how spatial data are produced and consumed have rapidly expanded to new users and platforms (Goodchild, Aubrecht, & Bhaduri, 2017; Sui, Elwood, & Goodchild, 2013).

The ubiquitous and accessible nature of big (geo)data facilitated a proliferation of non-traditional spatial data products and innovative data collection methods in research exploring human-environment interactions (Longley & Adnan, 2016; See et al., 2016; Senaratne, Mobasher, Ali, Capineri, & Haklay, 2017). Mobile phones and other portable electronic devices provide individuals with sensors of their environments capable of generating and sharing large amounts of geospatial data. Digital footprints, or the “digital traces each of us leaves behind as we conduct our lives,” (Weaver & Gahegan, 2007, p. 324), can be considered as a form of volunteered geographic information (VGI). Digital footprints captured by mobile devices and shared through social media platforms, like geotagged photos, offer an opportunity to study human-environment interactions central to geography at multiple spatial and temporal resolutions (Wang, Guo, Fu, & Li, 2014). This research explores VGI originating outside of formal research endeavors to quantify human presence and model infrastructure and environment interactions in areas designated for biodiversity conservation.

Protected areas (PAs) are designated terrestrial and marine

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landscapes formally managed to conserve nature, ecosystem services, and cultural values. PAs are core components in the strategy to minimize biodiversity loss globally, and human (i.e., visitor) data are often limited due to myriad competing research needs and challenges associated with visitor data collection. Sampling difficulties and time intensive methods in natural areas can serve as barriers to detailed visitor data (Cessford & Muhar, 2003), especially in large or topographically complex PAs. Additionally, staffing or financial constraints may necessitate prioritizing other objectives or concentrating visitor monitoring to select locations.

The dearth of visitor data in PAs is credited as a main limitation in implementing proactive management strategies to minimize visitor impact on resources (Hadwen, Hill, & Pickering, 2008) and to measure ecosystem services (Schägnier, Brander, Maes, Paracchini, & Hartje, 2016). Encroaching development or resource extraction near PA boundaries, climate change, and limited budgets can challenge efforts to balance use with biodiversity. To understand the spatial implications of environmental changes on visitor use patterns and support sustainable and effective management of PAs, managers require timely data on resource conditions and visitor use at site specific and landscape scales, as well as analytical tools to conduct integrated analyses to understand human-environment (i.e., visitor-environment) interactions.

This study contributes to the spatial-temporal understanding of human visitation and distribution patterns in PAs by leveraging digital footprints, manifested as publicly shared geotagged photos, as an efficient spatial data source to (1) model spatial distributions of visitor use to identify patterns of visitation at multiple temporal scales and (2) assess relationships between infrastructure, environmental factors, and visitor distribution patterns. The novel combination of geotagged photos and machine learning can identify annual and seasonal importance of infrastructure and environmental factors and inform management strategies for conserving natural resources, while providing opportunities for tourism and recreation.

2. Literature review

2.1. Spatial visitor use monitoring

Visitor use and impact monitoring within PAs focuses on use levels, locations, and associated environmental and social impacts to identify trends and alert management to departures from desired conditions (Cessford & Burns, 2008; Leung & Monz, 2006; Tomczyk & Ewertowski, 2011). The number of visitors to an area represents the most basic yet vital information for PAs, important for planning facility design and capacity, aligning with PA missions, and identifying potential for negative impacts (Cessford & Muhar, 2003). A variety of counting methods exist, ranging from observation studies, registries, permits, and on-site surveys to the use of electronic traffic counters, sensors, and photographic or videographic methods (Arnberger, Haider, & Brandenburg, 2005; Marvin et al., 2016; O'Connor, Zerger, & Itami, 2005; Xia & Arrowsmith, 2008). Global positioning system (GPS) tracking and computer modelling represent increasingly leveraged methods for understanding spatial use patterns and provide high-resolution data to quantify use intensities, explore changes in spatial-temporal patterns, and understand resource impacts (Beeco & Brown, 2013; Hallo et al., 2012; Meijles, de Bakker, Groote, & Barske, 2014; Van Kirk, Martin, Ross, & Douglas, 2014). Collectively, traditional and emerging approaches provide necessary and important information for PA conservation. Cost, efficiency, or the environment can still hinder wider or repeat implementation (Eagles, 2014). Additionally, data collection is often the responsibility of the research partner or managing organization and subject to staffing and budget constraints.

Sampling strategies prioritizing high use areas or peak visitation periods are vital in protecting ecological and site integrity at well-known destinations. However, this approach may miss emerging trends or seasonal variations. Emerging trends can lead to unintended negative impacts. Recreation ecology literature has shown that impacts can happen

quickly, with small amounts of use causing disproportionate amounts of impact (Hammit, Cole, & Monz, 2015; Monz, Cole, Leung, & Marion, 2010). Furthermore, although changing climate patterns vary geographically in intensity and manifestations, changes in the location or quality of resources may affect utilization and visitor-environment interaction patterns (Jones & Scott, 2006). Recent studies demonstrate that visitors are already changing their decisions about the timing, frequency, or duration of PA visits (Buckley & Foushee, 2012; Loomis & Richardson, 2006; McEvoy, Cavan, Handley, McMorrow, & Lindley, 2008; Richardson & Loomis, 2005; Scott, Jones, & Konopek, 2007), further complicating the sampling schemes of visitor data collection.

Temporal changes in visitation patterns, coupled with changing environmental conditions, also suggests the possibility of spatial shifts in visitor-environment interactions. Changes in average precipitation and temperature have altered some species distributions (Chen, Hill, Ohlemüller, Roy, & Thomas, 2011) and phenology (Lesica & Kittelson, 2010; Menzel et al., 2006), together with changing seasonal access in PAs, affords opportunities for temporal and spatial changes in use patterns. These changing patterns may not be congruent with existing infrastructure (i.e., managed trails) or management strategies, resulting in new or increasing negative impacts. Quantifying and visualizing visitor distributions is especially relevant when the landscape itself is also dynamic or unique, as patterns may change quickly or unexpectedly. Therefore, methods are also needed to collect visitor use data at a landscape scale without unduly burdening data collection and analysis efforts.

2.2. Research applications of geotagged photographs

VGI has helped address data collection constraints, with increasing work on accuracy, reliability, and credibility (Connors, Lei, & Kelly, 2012; Flanagan & Metzger, 2008; Levin, Lechner, & Brown, 2017; Senaratne et al., 2017). Including the public in scientific data collection (citizen science) continues as an active area of research in geography and GIScience, though less attention has focused on crowdsourced data arising outside of a formal research collaboration or citizen science effort (Connors et al., 2012). These platforms offer a potential complement to informing monitoring efforts of emerging trends or time-sensitive spatial data needs beyond the capacity of traditional monitoring.

Photo sharing sites like Flickr, Panoramio, Instagram, and others allow for cloud storage of user photos and map-based visualization of geotagged photo locations (van Zanten et al., 2016). In accordance with a site's privacy and use policy, researchers can query photo metadata using the site's application programming interface (API). APIs allow requests to be made of a server, app, or software, which then responds with data. The use of APIs to extract human movement patterns from geotagged media has progressively been documented in tourism and recreation literature (see Zheng, Zha, & Chua, 2010), with Flickr offering a popular platform due to the number of photos and accessible API (Levin et al., 2017). By 2007, one year after the inclusion of geotagging functionality, Flickr hosted over 20 million geotagged photos (Zheng et al., 2010). That number does not take into account the large volume of photos without geotags, for which researchers are also developing processes to assign spatial attributes (Kalogerakis, Vesselova, Hays, Efros, & Hertzmann, 2009; Liu, Yuan, Cong, & Xu, 2014).

Research also suggests that Flickr data can be spatially accurate (Zandbergen & Barbeau, 2011; Zielstra & Hochmair, 2013) and timely (Antonioni, Morley, & Haklay, 2010). Flickr users have the option of manually geotagging a photo by placing it on a map if the camera or device does not have internal GPS capabilities, or leveraging an external GPS device to record the geographic location (Senaratne et al., 2017). Of the photos uploaded to Flickr in 2014, the most common devices included popular smart phone brands (Dove, 2015). Mobile phones with GPS are capable of achieving horizontal accuracy similar to recreational grade GPS units, often within 10 m of true position (Zandbergen & Barbeau, 2011).

The volume of geotagged photos suggests the impact of incorrectly

tagged photos can be minimal, and crowdsourced correction capacity exists on sharing sites. Zielstra and Hochmair (2013) examined the spatial accuracy of geotagged photos through a random selection of geotagged photos from Flickr and Panoramio. Their results indicated median positional errors of 15 m and 46 m in North America for Panoramio and Flickr images, respectively, when compared to manually corrected position based on image content. Antoniou et al. (2010) found geotagged photos in the UK were usually also uploaded within a few weeks following capture. In their sample, only 8.4% of Flickr photos were uploaded more than a year later, suggesting that data from Flickr can be both accurate and timely enough for some applications.

Geotagged photo data related to visitors in PAs have found the number of photos uploaded were positively correlated with other visitor monitoring methods (Sonter et al., 2016; Wood, Guerry, Silver, & Lacayo, 2013). Wood et al. (2013) used data from Flickr to estimate visitation rates at 836 natural and cultural recreation sites in 31 countries and found the number of uploaded photos was positively correlated with empirical visitation counts. Sonter et al. (2016) also found correlations between photo numbers and survey visits to PAs in Vermont, USA, and estimated economic contribution of nature-based tourism and landscape characteristics influencing the spatial-temporal patterns observed using Flickr photos.

Beyond counts, Flickr photos also have the potential to document visitor movements and distribution within a PA. Addressing concerns about the relatively low density of geotagged photos in natural areas, Orsi and Geneletti (2013) employed a gravity model to estimate visitor flows for the Dolomites UNESCO World Heritage Site based on popular locations determined through high densities of geotagged photos validated with on-site counts. García-Palomares, Gutiérrez, and Mínguez (2015) leveraged photos from Panoramio to identify hotspots within European cities, differentiating between local residents and tourists. Results highlight how geotagged photos can provide use estimates and hot spots for large areas in a quick and inexpensive way.

Counts and estimated volumes are critical first steps toward understanding potential visitor pressure on natural resources within a PA, with research also needed to address what variables are influencing such distributions of use. Richards and Friess (2015) classified the content of geotagged photos to examine cultural ecosystem services (CES) in natural areas of Singapore. Using geotagged photos provided a rapid assessment of what visitors derived from different habitats, as well as broad activity categories. Oteros-Rozas, Martín-López, Fagerholm, Bieling, and Plieninger (2017) also conducted a content analysis of geotagged photos and found that recreation and cultural heritage were the most common CES photographed across five European sites. Their study also found that Flickr, when compared with Panoramio, included more CES photos. Keeler et al. (2015) used geotagged photos as a proxy for recreational visits to Midwestern lakes in the U.S. with results indicating increased visits corresponded with increased water clarity.

The accuracy, timeliness, and availability of geotagged photos represents a complement to traditional visitor use monitoring methods to assist in gathering park to landscape scale data to inform focused monitoring efforts and possible resource impacts. The ability of geotagged photos to visualize visitor distributions and potentially identify human-environment interactions difficult to capture with other methods is especially relevant when the landscape itself is also dynamic.

3. Methods

3.1. Study area

Hawaii Volcanoes National Park (HVNPN) is an example of a frequently changing landscape in which data on human-environment interactions offer a unique opportunity to explore spatial-temporal changes in visitor use patterns. HVNPN is located on the Island of Hawaii, the southernmost island in the Hawaiian Archipelago (Fig. 1), and managed by the U.S. National Park Service (NPS). HVNPN is also a

United Nations Educational, Scientific and Cultural Organization (UNESCO) Biosphere Reserve (UNESCO, 2016) and UNESCO World Heritage site. Both designations make HVNPN an important component of global conservation efforts, geared toward innovative and integrative strategies to protect outstanding natural and cultural values.

Due to the geographically isolated location of the archipelago, HVNPN conserves habitat for a multitude of endemic plant and animal species, including 54 threatened or endangered species within seven contiguous biotic zones in the park's nearly 1300 km² (IUCN, 1987). Mauna Loa, considered the largest volcano in the world, is one of two active shield volcanoes in the park. Kilauea is the second volcano in the park, erupting continuously since 1983 and adding over 200 ha of new land and covering 14 km of the Chain of Craters Road along the southern boundary of the park.

The actively changing landscape, unique volcanic activity, geologic features, and cultural importance attracted over 1.8 million visits in 2015 and 2016 (NPS, 2016), and was ranked as the third most visited attraction, natural or built, in the Hawaiian Islands in 2015 (Hawaii Tourism Authority, 2015). The level of visitation and rapidly changing landscape of HVNPN underscores the need for efficient snapshots of visitor use distributions. The potential for new visitor use patterns and locations, especially relevant in an actively changing landscape, could be supported through an understanding of emerging trends in the spatial distribution of visitors available from novel data sources like geotagged photos.

3.2. Geotagged photo acquisition

Geotagged photos were queried from the Flickr API using the statistical software program R (version 3.2.5). We chose Flickr because it has been included as a data source in GIScience, geography, and tourism literature, offers an accessible API, and has experienced continued growth (Alivand & Hochmair, 2017). Using the boundary of HVNPN, we collected attributes on all publicly shared photos containing a geotagged taken between January 1, 2011 and December 31, 2015. Attributes included latitude, longitude, owner, date taken, date uploaded, number of views, text tags, photo title, accuracy rating, image URL, and capture device. Records were analyzed in R and converted to shapefiles for geospatial analysis using ESRI ArcGIS (10.3.1).

As different resource management questions focus on multiple levels and locations of use, each photo was considered an incident of use and not weighted for multiple uploads by the same individual. If the same individual uploaded multiple photos from different locations within the park, each new spatial location is an indicator of use. Similarly, if an individual uploaded multiple photos from the same location taken over the course of several hours, the length of stay has resource use and impact implications. For example, visitor presence may displace wildlife from that location, indicating that the duration, and therefore each incidence, is important.

The content of photos was also assessed to gauge the level of spatial accuracy. Photos within 50 m of known or identifiable landmarks (e.g., Visitor Center, popular trails, campsites, coast, iconic craters and calderas) were identified in ArcGIS. From those photos, a random sample of 10% were selected for manual review. If a photo was no longer available, or zoomed to a level where the landmark was difficult to determine, the title and tags were checked for confirmation.

3.3. Visitor spatial distribution modelling

Distribution modelling assumes that the phenomenon of interest will occur in locations with suitable conditions, that a systematic assessment of absence has been conducted, and that observations are independent. Geotagged photos violate these assumptions, as do many plant and animal species distribution data sources. Maximum entropy (MaxEnt) models have been successfully applied using incomplete and correlated information (Elith et al., 2006, 2011). Used widely in species distribution modelling in ecology (Phillips, Anderson, & Schapire, 2006), MaxEnt has

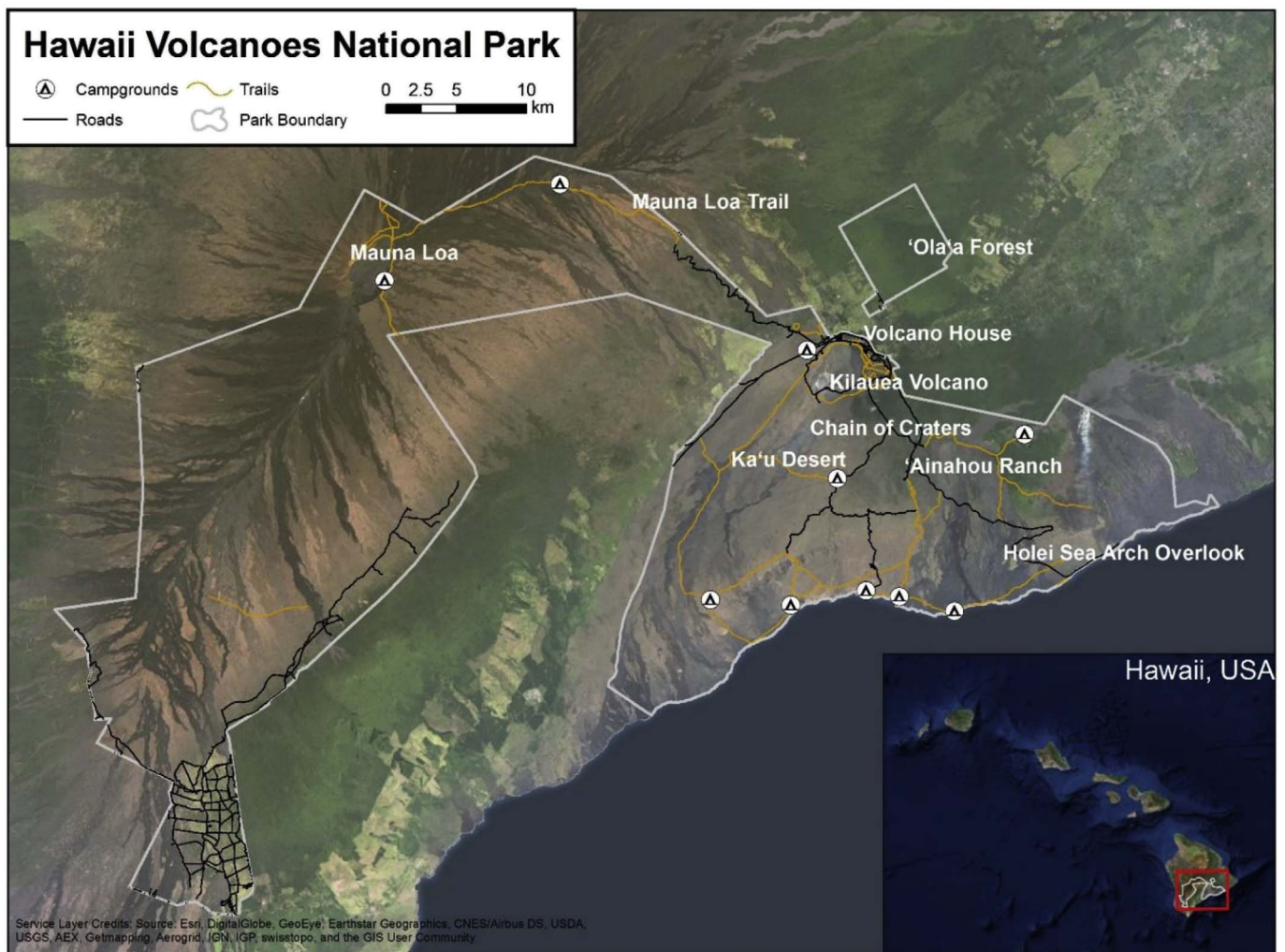


Fig. 1. Map of Hawaii Volcanoes National Park.

only recently been applied to visitor use distributions in tourism and recreation research (Westcott & Andrew, 2015). MaxEnt models integrate continuous and categorical predictor variables, minimize overfitting, and evaluate each covariate's influence. While other distribution modelling approaches can achieve similar objectives, MaxEnt was chosen because it has been shown to perform well, if not better, in comparison to other approaches when absence data are unavailable (Braunisch & Suchant, 2010; Elith et al., 2006) and offers an accessible graphical user interface. The use of MaxEnt for PA management of visitor impacts on natural resources has included human-wildlife interactions (Braunisch, Patthey, & Arlettaz, 2011), prediction of off-trail recreation behavior (Coppes & Braunisch, 2013; Westcott & Andrew, 2015), and cultural ecosystem services (Richards & Friess, 2015).

Recent critiques of MaxEnt have included explorations of mathematical similarities to Poisson point process models (PPM) (Renner & Warton, 2013). Equivalencies between the two methods depend on how model parameters are configured and analyzed. Renner et al. (2015) demonstrated how MaxEnt could be used as a means for fitting a Poisson PPM, namely by retaining duplicate presence records and examining the effect of the number of background points on the regularized training gain. Additionally, the influence and handling of spatial autocorrelation in model residuals remains an open question concerning MaxEnt modelling (Halvorsen et al., 2016).

For the purposes of this analysis, we focused on distributions within a park rather than predicting patterns in new environments. Duplicate presence points were retained and the MaxEnt algorithm chose the best feature type (i.e., linear, quadratic, product, threshold, or hinge). In

machine learning, features are transformations of original covariate data, similar to basis functions in regression approaches, used to model complex, non-linear responses (Elith et al., 2011). Models presented in this paper were based on 10 replications for each temporal segment of interest, using random seed subsampling and withholding 30% of the data for training. The maximum number of background points was set at 10,000 with a convergence threshold of 0.00001 (Merow, Smith, & Silander, 2013; Phillips et al., 2006; Poor, Loucks, Jakes, & Urban, 2012; Young et al., 2011).

3.4. Independent variable data sources

Visitor infrastructure (i.e., distance to buildings, parking, roads, trails, and campsites) and environmental characteristics (i.e., vegetation type, elevation, slope, and distance to water) served as independent variables in the MaxEnt model. Infrastructure variables were chosen based on their purpose to serve as either transportation corridors or destinations. Additionally, these factors are components of management plans and represent ways agencies provide visitor services as well as mitigate impacts. The environmental variables were chosen based on recreation ecology literature suggesting sensitivity of visitor use (e.g., trails on steep slopes subject to erosion, some vegetation types – including in higher elevation areas – are less resistant or resilient to trampling) as well as known environmental features attracting use (Hammit et al., 2015). In cases where data for these physical attributes was available in point, line, or polygon representations, the distance to each feature was calculated and resulting values were converted to

raster. Using linear distance served to quantify the position relative to the attributes of interest. All independent variables either were originally, or resampled to, the coarsest resolution (i.e. 30 m). Response curves for each predictor were created and the variables ranked based on their overall contribution to the model. Response curves illustrate how values of the predictor variable influence the logistic probability of presence, with upward trends indicating positive associations. The curves are separate MaxEnt models with only the predictor variable of interest and clamped at the minimum and maximum values identified in the training data to minimize errors associated with extrapolation.

Predictive accuracy was evaluated using the area under the curve (AUC) (Jalali, Ierodiaconou, Monk, Gorfine, & Rattray, 2015). Scores greater than 0.75 are considered acceptable for predictive accuracy (Phillips & Dudík, 2008; Richards & Friess, 2015), with AUC scores greater than 0.9 considered excellent (Coppes & Braunisch, 2013). The influence of each predictor variable was examined through permutation importance because the values are not dependent on the path taken by the model to arrive at a solution. Permutation importance reevaluates the MaxEnt model by randomizing the values for each variable separately to measure the decrease in AUC.

Distribution model interpretations and results were compared with zonal statistics within 1 km of known features and statistically significant hot spots as identified using the Optimized Getis-Ord G_i^* statistic. As a local indicator of spatial association (LISA), the Getis-Ord G_i^* statistic provides an additional level of analysis beyond density maps and distribution models to quantify and visualize locations of high or low density, reducing subjectivity in assessing the resulting visualization by identifying only locations achieving statistical significance. Additionally, for the seasonal models, the raster calculator examined the difference between summer and winter annual means to visualize the difference.

4. Results

4.1. Geotagged photo sample characteristics

A query of public, geotagged photos on Flickr taken in the park between January 1, 2011 and December 31, 2015 resulted in 12,048 records uploaded by 897 Flickr users, with a median of four photo uploads per user (interquartile range: 1 to 12 photos per user). Photos were uploaded to Flickr a median 12 days after capture (interquartile range: 2.75–35 days), with 40% of the sample uploaded within a week and increasing to over 70% within 30 days. Devices used to capture photos included film (0.18%) and digital cameras (66.4%), as well as mobile devices (19.3%). Although digital cameras comprised more than two-thirds of the total 5-year sample (66.4%), recent years indicated an increasing proportion of mobile devices (Fig. 2).

Photos within 50 m of known landmarks were identified. This included 4752 photos, 10% (475) of which were randomly selected for manual review. Of those 475, only 3 (0.63%) were no longer viewable,

Table 1

Number of geotagged photos posted to Flickr within park boundary by season and year.

	Winter (Dec – Feb)	Spring (Mar – May)	Summer (Jun – Aug)	Fall (Sep – Nov)	Total
2011	428	680	649	703	2460
2012	539	454	414	930	2337
2013	474	747	1324	326	2871
2014	639	412	1227	492	2770
2015	521	370	393	326	1610
Total	2601	2663	4007	2777	12,048

but the title and tags confirmed the location. For close-up photos ($n = 17$) where the landmark was difficult to see, 16 included birds and plants found in the park and confirmed by the title and tags. The remaining close up was a selfie with a title that referenced the correct location. Only four photos ($n = 0.84\%$) were potentially incorrect. These included what appeared to be historical photos ($n = 2$) or potentially taken outside of the park ($n = 2$). As it was not possible to determine if the historical images where photos taken at an interpretive exhibit or scanned documents, they were retained for the analysis. Similarly, it was difficult to determine if the two remaining photos were indeed taken outside of the park, and therefore were retained for analysis as the multiple, random samples used in the distribution models minimized the influence of spatial outliers, which comprised less than half of one percent of the sample.

Records were grouped by year and season to investigate annual and seasonal variation in the models, respectively (Table 1). Years 2013 and 2014 had the greatest number of photos with over 2700 taken each year. A smaller sample size of approximately 1600 was obtained for 2015. Aggregated by season, the greatest number of photos taken and shared varied by year, with the fall (i.e., September through November) having the highest number of photos in 2011 and 2012, and shifting to the summer months (i.e., June through August) in 2013 and 2014. In 2015, winter months (i.e., December through February) witnessed the largest sample size.

4.2. MaxEnt models

4.2.1. Annual models

Visualizations of model output across years show similar locations with high logistic probabilities of visitor presence, spatially coinciding with roads, trails, and other visitor infrastructure (Figs. 1 and 3). Few differences were visually identifiable between the years, with the exception of 2014. In 2014, increased probability of presence was observed along the northern portion of the park (i.e., Mauna Loa trail and summit) compared to other years. Corroborating the visual assessment, we used zonal statistics to examine MaxEnt values within a 1 km radius of the Mauna Loa trailhead for each year. The 2014 model indicated the

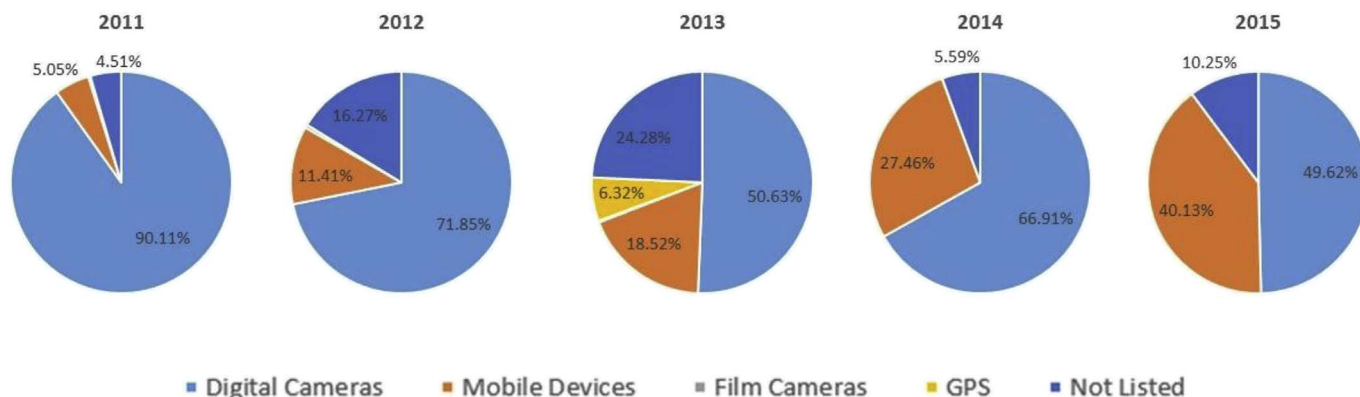


Fig. 2. Type of device used to capture image shared to Flickr. Percentages based on the total number of records per year.

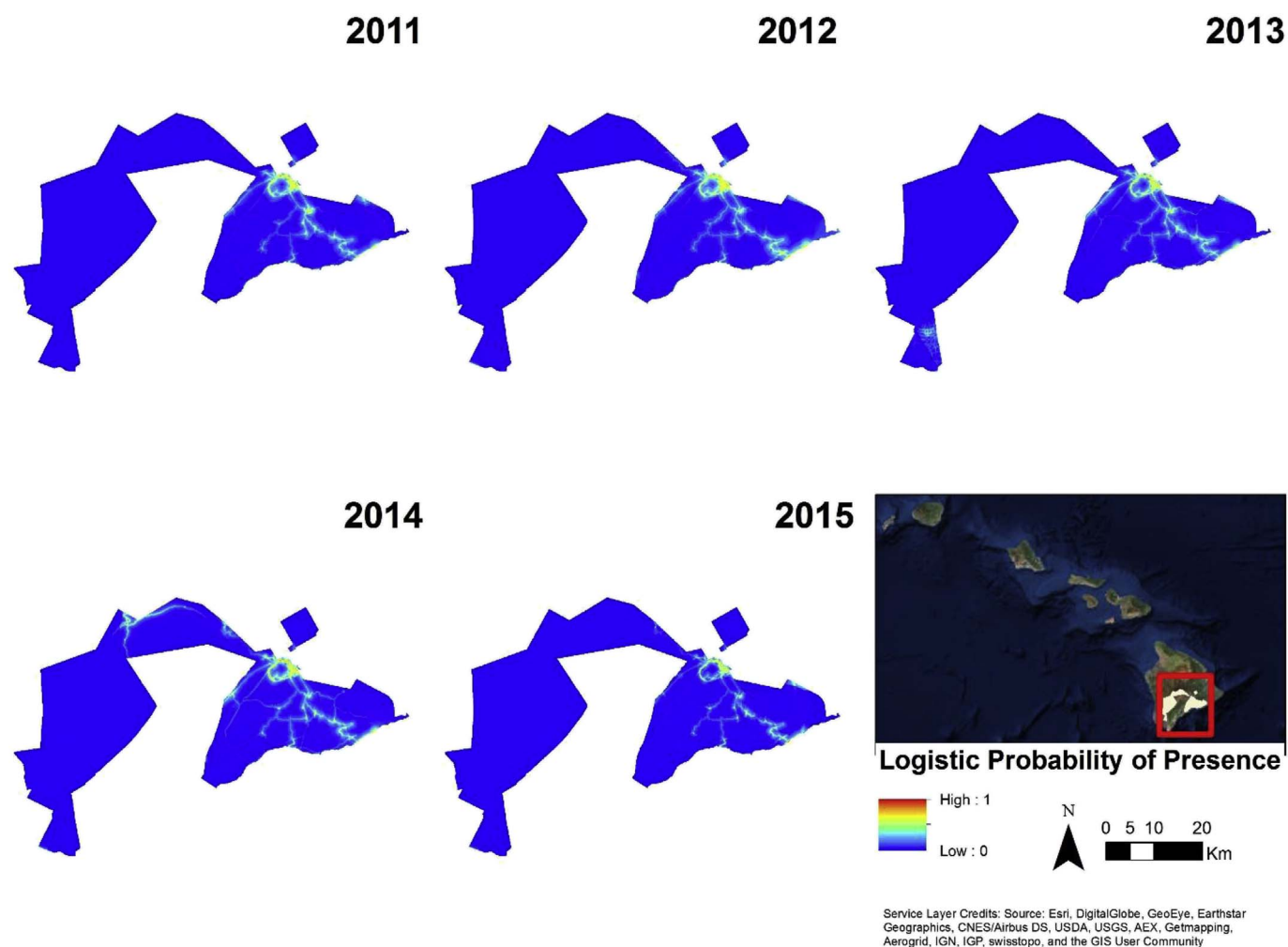


Fig. 3. MaxEnt models by year. Warmer colors indicate higher logistic probabilities of presence. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

highest logistic probability of presence values when compared to the same area across years (Table S1, Supplementary material).

Consistent across years was an increased logistic probability of presence in the northeastern part of the park, coincident with the roads and trails providing access to the park's visitor center, hotel (i.e., Volcano House), and the Kilauea caldera. Maximum logistic probability of presence values were above 0.8 for all years within a 1 km radius of the park's hotel (Table S1 in supplementary materials). Examining the concentrated area of high probability of use near the northeastern part of the park (Fig. 4), similar hot spots (i.e., orange to red) persist, with less to the west of the caldera in 2012 and 2015. These findings were also supported by Getis-Ord G_i^* statistics, which identified several significant visitor hot spots concentrated along roads and trails near the park's visitor center, hotel, and main volcanic craters and calderas (Fig. S1 and S2, Supplementary material).

For all years, distance to a road was a top contributing variable, representing at least 30% of AUC training gain (Table 2). It was the top contributing variable in all models with the exception of 2014, when distance to trail was the top contributing variable at 43.2%. Distance to trail also contributed at least 20% to the overall model in 2014 and 2015. In 2012 and 2013, elevation also contributed at least 20% of the AUC training gain. For all six models, the predictive accuracy was considered high with AUC values greater than 0.9.

To exercise caution when interpreting variable contribution due to potential correlation among predictor variables, variables were also examined as individual response curves. Distance to roads and trails

indicated high logistic probability of presence on or within a few hundred meters of the infrastructure (Fig. 5). Similar response curve profiles resulted for distance to road, with the exception of 2014. In 2014, an increasing logistic probability of presence was identified as distance increased above 17,500 m from the road. This corresponds to increased logistic probability of presence along the trail to the summit of Mauna Loa (see Fig. 3), which is over 27 km from where the park road ends at the trailhead. Higher logistic probability of visitor presence was also observed on or very near trails, dropping sharply after a few hundred meters.

4.2.2. Seasonal models

Results from seasonal models also highlight increased logistic probability of presence near the visitor center and Kilauea caldera, as well as along Chain of Craters Road (Fig. 6). More noticeable in the seasonal models are differences in visitor presence along Mauna Loa trail in the summer months, with greatest logistic probability of presence values within 1 km of the trailhead (Table S2 in supplementary materials), and lower logistic probability of presence on trails in the Kua Desert in the winter and fall as compared to the spring and summer months (Figs. 1 and 6). These findings were confirmed by examining the difference between summer and winter annual means using the raster calculator. Visualizations of the difference are included in the Supplementary material (Fig. S4). Lower logistic probability of presence along trails was also noted during the fall, with greater values concentrated along the roads in the southern portion of the park.

Model visualizations near the visitor center and Kilauea caldera

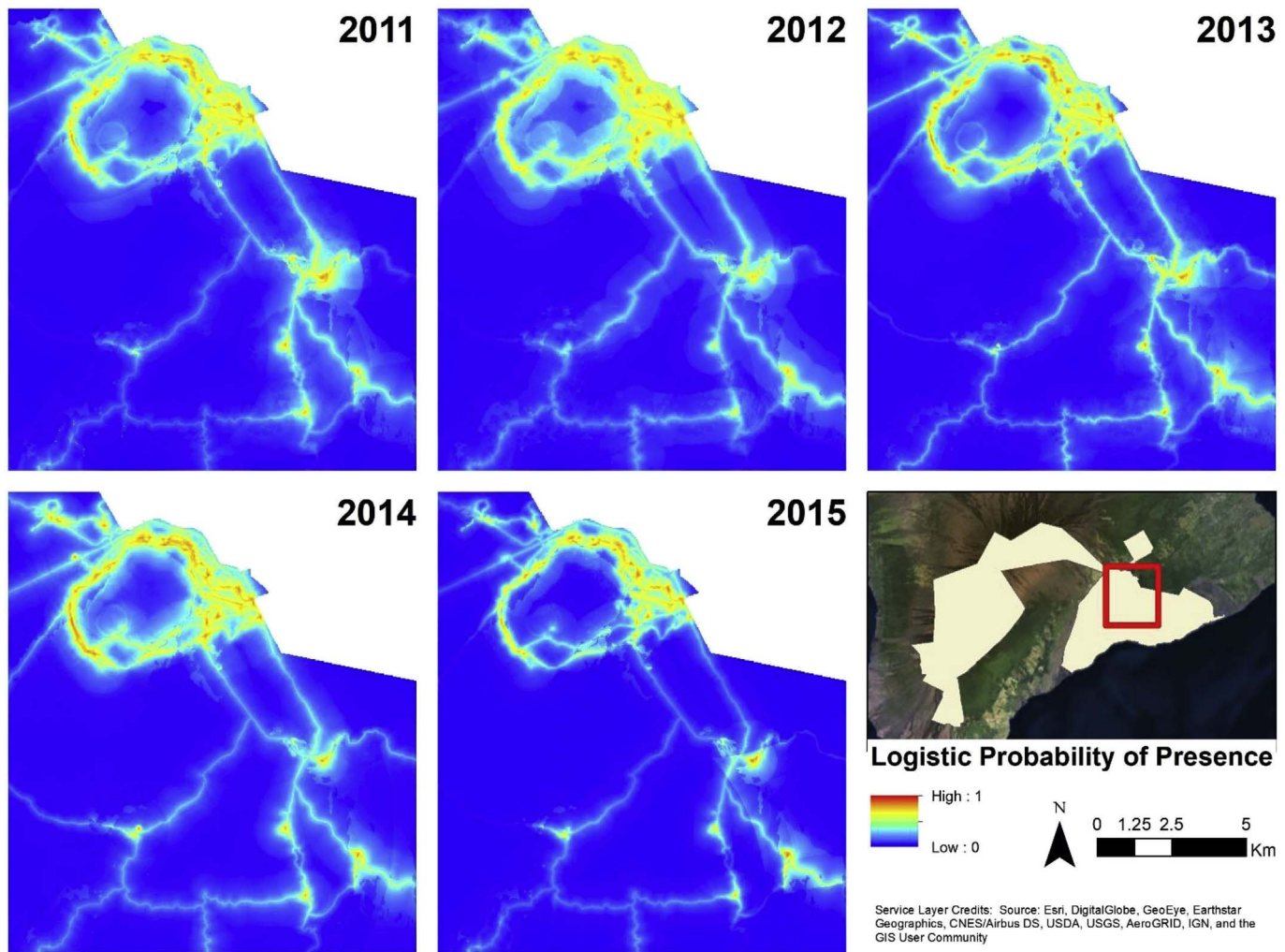


Fig. 4. MaxEnt model outputs by year near park visitor center and Kilauea caldera.

Table 2

Permutated importance for environmental and managerial variables by year. Bolded values indicate the variable contributed at least 20% to the AUC training gain.

	Permutated Importance (%)					
	2011	2012	2013	2014	2015	All Years
Distance to Road	44.3	41.7	41.2	30.4	57.1	37.8
Distance to Parking	10.9	15.1	1.9	13.4	3.2	2.8
Distance to Trail	11.1	5.1	8.4	43.2	21	16
Distance to Building	12.1	1.6	10.4	2	6.1	12
Elevation	18.1	32.7	34.4	4.8	9.9	24.9
Distance to Campsite	1.3	2.1	1.8	1.9	0.4	3.1
Land Cover	1.7	1.2	0.9	1.3	1.1	1.1
Distance to Water	0.5	0.3	0.8	2.9	0.9	2.1
Slope	0.1	0.1	0.2	0.1	0.2	0.2
Test AUC	0.960	0.960	0.958	0.957	0.973	0.912
AUC St. Dev.	0.005	0.003	0.003	0.003	0.004	0.003

indicate more areas of very high logistic probability of presence (i.e., orange and red) during the summer months along the eastern edge of the caldera and along the western edge during the winter (Fig. 7). These hot spots of use (as noted in orange and red) also occurred to a greater linear extent during the summer. Visualizations of spring and fall probabilities of presence appear similar, with high probabilities of presence along the trail network and near the visitor center.

As in the annual models, the top contributing predictor variables to the model included distance to roads for all seasons (Table 3). Distance

to trail contributed more than 37% in the summer months, and elevation contributed more than 54% in the spring. Distance to trail did not contribute as much in the seasonal models, with the exception of the summer, as compared to annual models. This finding is also evident in the model visualizations, with visitor probability of presence concentrated on roads and less on trails in the fall and winter (Fig. 6). Predictive accuracy for the four seasonal models was high, with AUC values greater than 0.9.

Response curves for the seasonal models reflected findings in the annual models, with visitor presence very likely on or within a few hundred meters of a road or trail, with an increase in the summer months to approximately 15,000 m, related to the increased logistic probability of presence along the Mauna Loa Trail and summit. The response curves for elevation indicate high logistic probability of presence at sea level and 1000–1250 m above sea level (a.s.l.), the latter corresponding to the elevations of the visitor center and hotel near Kilauea caldera. Increased logistic probability of presence above 3000 m a.s.l. during the summer months, also related to increased probability of presence along Mauna Loa Trail, which starts at approximately 2000 m a.s.l and ends just above 4000 m a.s.l.

5. Discussion

Flickr photo locations provided data on the spatial and temporal patterns of visitation across the park. Specifically, inferences derived from MaxEnt models highlight the importance of infrastructure in the overall distribution of visitor photos, similar to MaxEnt models of

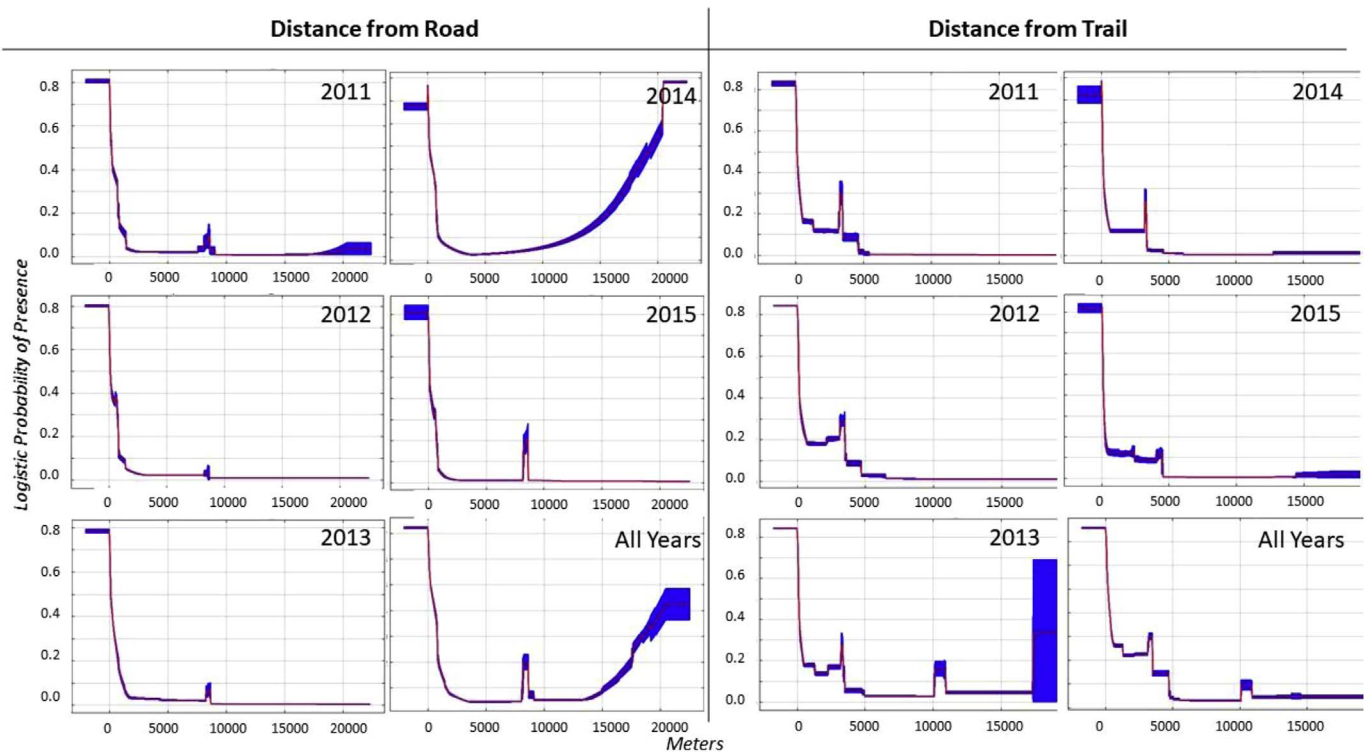


Fig. 5. MaxEnt response curves for distance to roads and trails by year. The red line represents the logistic probability of presence, with the blue shading providing a plus and minus one standard deviation from 10 replicate runs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

winter recreation impacts on wildlife studied by [Coppes and Braunisch \(2013\)](#). The high logistic probability of presence along roads is an important empirical finding for mitigating roadside impacts to vegetation, as research has found they can be greater and more extensive than trailside impacts ([Wolf & Croft, 2014](#)). It also underscores how infrastructure and transportation planning can influence potential impacts to flora and fauna ([Monz, D'Antonio, Lawson, Barber, & Newman, 2016](#)) based on when, where, and how areas are made accessible to visitors. Additionally, while the persistence of high probabilities of presence on or very near infrastructure supports confinement strategies to keep visitors safe and minimize resource impacts associated with off-

trail travel, the data highlight potential locations for social impacts to visitor experiences like perceived crowding, which could be measured through surveys and observations. These findings also help inform additional location or time-specific monitoring using complementary techniques, which can be paired with approaches to analyze visitor behavior ([Antoniou et al., 2010; Dunkel, 2015](#)).

Considering the spatial location of variable values in conjunction with the logistic probability of presence is important for understanding distribution patterns. Elevation, for example, which contributed over 20% to some of the models, had response curves that indicated increase logistic probability of presence between 1000 and 1400 m a.s.l., and

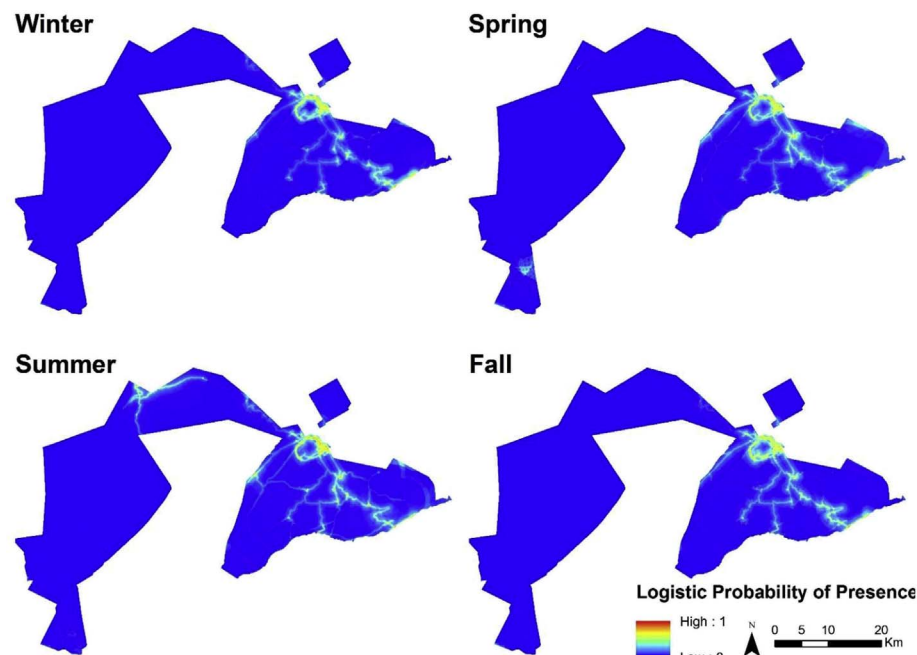


Fig. 6. MaxEnt models by season. Warmer colors indicate higher logistic probability of visitor presence as modeled using geotagged photo records. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

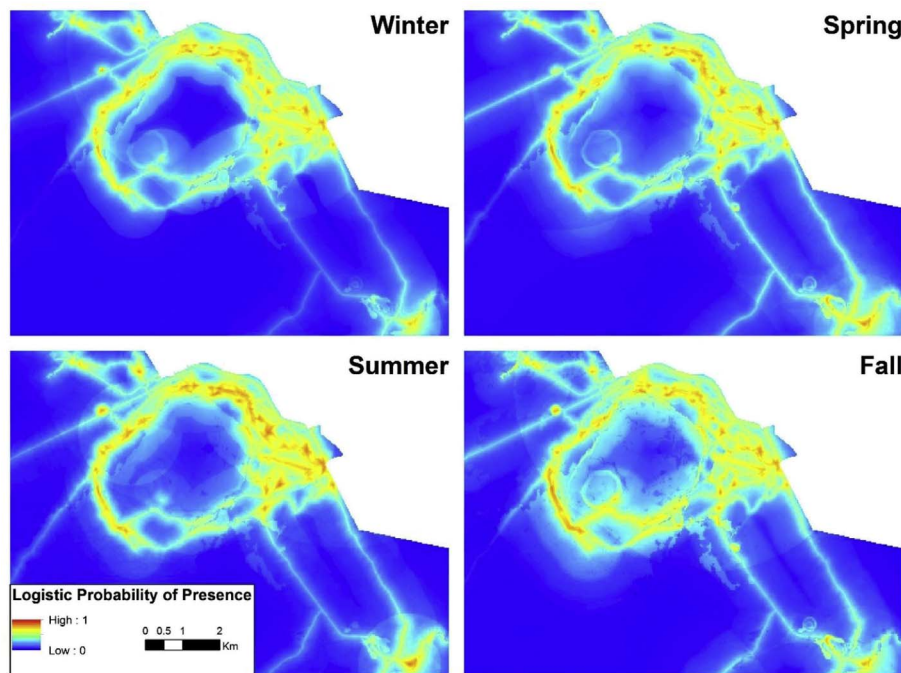


Fig. 7. MaxEnt model outputs by season near park visitor center and Kilauea caldera.

Table 3

Permutated importance for environmental and managerial variables by season. Bolded values represent variables contributing to at least 20% of the AUC gain.

	Permutated Importance (%)				
	Winter	Spring	Summer	Fall	All Seasons
Distance to Road	46	29.4	33.3	58.7	37.8
Distance to Parking	13.1	1.1	7.3	13.2	2.8
Distance to Trail	14.6	5.7	37.8	10.4	16
Distance to Building	9	6.3	8.6	2.9	12
Elevation	12.5	54.2	8.9	10.9	24.9
Distance to Campsite	1.8	1.4	2.6	0.9	3.1
Land Cover	1.1	0.9	0.7	2	1.1
Distance to Water	2	0.8	0.7	0.6	2.1
Slope	0.1	0.1	0.2	0.4	0.2
Test AUC	0.962	0.953	0.952	0.959	0.912
AUC St. Dev.	0.002	0.004	0.004	0.003	0.003

again above 3500 m a.s.l. Those elevations correspond to park infrastructure (e.g., Volcano House sits at approximately 1220 m a.s.l.) and the trail along Mauna Loa (with a starting elevation near 2000 m a.s.l., and ending at the summit just above 4000 m a.s.l.). The importance of elevation to overall spatial patterns also represents an attribute for further consideration in areas where visitor infrastructure traverses more elevation gradients and when exploring seasonal patterns. Specifically, explorations of areas with documented shifts in species presence, climate and weather patterns, and fire regimes would advance our understanding of if and how visitors are changing their spatial patterns of use (McEvoy et al., 2008). Additionally, further examining probabilities of presence by time of day could highlight important findings regarding the fluctuations of use throughout a day.

Surprising in the photo records and MaxEnt models was the high logistic probability of presence along the western part of Kilauea caldera (Figs. 4 and 7) as parts of this area closed in 2008 due to hazardous conditions. Random inspection of photos georeferenced to the area indicated many were aerial photographs, suggesting visitors were present in the area, however not participating in ground-based activity. Few appeared to be the result of incorrect geotags, which has been identified as a source of error in other studies (Orsi & Geneletti, 2013; Zielstra & Hochmair, 2013).

The finding that some photos were taken from aircraft is important to other studies assessing impacts based on visitor presence with geotagged photos, given that photos may not represent ground-based activity. These records are still an important indicator of impact as helicopter tours can disturb wildlife (Stockwell, Bateman, & Berger, 1991) and contribute to noise pollution (Pickering, Harrington, & Worboys, 2003). This is also true of cable cars, chair lifts, or other air-based transportation and tourism operations, where the photo location has an altitude component. We recommend further research continue exploring distribution models following a content analysis of the photos (see Oteros-Rozas et al., 2017; Richards & Friess, 2015), classified by visitor activity and landscape features, to model activity zones as the impacts to natural resources vary.

Data were uploaded within about two weeks of their capture, similar to the timeliness noted by Antoniou et al. (2010). Users also uploaded similar numbers of photos as previous research (Richards & Friess, 2015; Sonter et al., 2016). The shift from digital cameras toward mobile devices of the devices reported between 2011 and 2015 has implications for spatial accuracy assumptions in future modelling and research. Many of the digital cameras reported do not have integrated GPS, though from the data in this study, we could not distinguish the geotagging method (i.e., external GPS for camera or manual map placement). If location services are turned on for a mobile device, the image will contain the geotag from the phone's internal GPS and potentially reduce opportunities for errors introduced during manual geotagging. However, this also represents a challenge as data sets contain devices capable of varying resolution (Girardin, Blat, Calabrese, Dal Fiore, & Ratti, 2008). Spatial accuracy assessment could improve parameter settings for models and spatial statistics, including the ones in this study, which are sensitive to distance metrics for calculations. To minimize this limitation, MaxEnt was selected over other modelling approaches as it attempts to minimize the influence of spatial inaccuracies through multiple model iterations using different random samples of test and train data. However, it is important to note, that the influence of spatial autocorrelation in MaxEnt model residuals may warrant further investigation if being used as a standalone method. In this context, we propose leveraging geotagged photos and MaxEnt to inform and complement additional visitor use monitoring approaches.

A limitation associated with geotagged photos, in addition to potentially incorrect geotags, are that they represent a sample of those visitors

taking and sharing geotagged media, and therefore potentially biased to digital footprints. Growing work on the quality, accuracy, and contributors of VGI has found that more educated individuals (Crampton et al., 2013), and those employed in management, business, science, and arts, were more likely to create geotagged media on Flickr and Twitter (Li, Goodchild, & Xu, 2013). The implication for spatial distribution modelling is that leveraging only one platform may miss some use types and result in different distributions or growth patterns (Alivand & Hochmair, 2017).

While recent estimates suggest that 11% of the 2.6 million geotagged photos on Flickr fall within protected areas around the world (Levin, Kark, & Crandall, 2015), suggesting the global applicability and replicability of this approach using Flickr, it is not limited to one social media platform. An example includes the globally-popular Instagram, with real-time uploads and an estimated 28% of individuals with internet access in the U.S. contributing content (Pew Research Center, 2016; van Zanten et al., 2016). More research is also needed to explore user demographics for different social media platforms and how experience, values, accessibility, and environmental characteristics influence if and at which locations visitors take and share photos to inform future modelling efforts (Levin et al., 2017). Additionally, pairing modelling results with a media content analysis at finer temporal scales may also illuminate how events, both natural and programmatic, influence the number, location, and timing of visitor photos. For example, the NPS along with media outlets published several stories throughout 2014 about the increased seismic activity from Mauna Loa similar to its last eruption in 1984 (Klemetti, 2014; USGS, 2014). Although a formal assessment of these publications was beyond the scope of this study, it may relate to the greater number of photos shared in 2014, especially the increased digital footprint along the Mauna Loa trail.

6. Conclusion

Crowdsourced geographic information can reduce barriers to obtaining spatially explicit data about visitation patterns in PAs to answer long-standing and emerging questions on human-environment interactions and, subsequently, inform planning and management efforts through GIScience. Geotagged photos are only one source of crowdsourced geographic information. In recent years, platforms including Strava, GPSies, Wikiloc and MapMyRide have also been leveraged to gather GPS tracks of a user's entire trip to answer management questions (Campelo & Nogueira Mendes, 2016; Korpilo, Virtanen, & Lehtävirta, 2017). Geotagged photos from Flickr illustrated temporal changes in the spatial patterns of visitor use, and how infrastructure and natural environmental characteristics were associated with distribution patterns. Patterns derived from MaxEnt models and hot spot analyses indicated a level of predictability in visitor use. This predictability is useful if existing empirical evidence can inform impact, interactions, and spatial analyses.

However, within the context of a changing climate and dynamic responses of ecosystems, this predictability may only hold at certain spatial and temporal scales and warrant further investigation of the role of temporal scale. An example includes analysis of visitor patterns coinciding with media coverage of events like visibility of migratory wildlife (i.e., whale watching, birding) or geological events (i.e., volcanic activity). This is possible with the availability of free and timely crowdsourced data, capable of high spatial resolution, and from sources experiencing growth in the amount and extent of content shared (Alivand & Hochmair, 2017). Continued exploration of these sources and their capacity for addressing conservation and PA management challenges represent a rich area of research in GIScience, as the supporting technologies and possibilities continue advancing.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.apgeog.2017.11.004>.

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